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Dynamic proportion portfolio insurance using genetic programming with principal component analysis ☆

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Abstract

This paper proposes a dynamic proportion portfolio insurance (DPPI) strategy based on the popular constant proportion portfolio insurance (CPPI) strategy. The constant multiplier in CPPI is generally regarded as the risk multiplier. Since the market changes constantly, we think that the risk multiplier should change according to market conditions. This research identifies risk variables relating to market conditions. These risk variables are used to build the equation tree for the risk multiplier by genetic programming. Experimental results show that our DPPI strategy is more profitable than traditional CPPI strategy. In addition, principal component analysis of the risk variables in equation trees indicates that among all the risk variables, risk-free interest rate influences the risk multiplier most.

Keywords: Dynamic proportion portfolio insurance (DPPI); Constant proportion portfolio insurance (CPPI); Genetic programming (GP); Principal component analysis (PCA)

1. Introduction

Constant proportion portfolio insurance (CPPI) is a simple strategy based on the investor's preference of risk to calculate the amount invested in the risky asset (Black & Jones, 1987; Black & Perold, 1992). The rest of capital is invested in the risk-free asset. The purpose of CPPI is to help investors capture the upside potential of the equity market while maintaining a minimum floor of the portfolio.

Because the risk multiplier in the CPPI strategy is predetermined by an investor's own view, inappropriate expectation of the market will result in loss (Hakano, Kopprasch, & Roman, 1989). When the market becomes more volatile, a larger risk multiplier may cause a larger loss. Thus, a new dynamic hedging model needs to be developed (Zhu & Kavee, 1988). Although the concept of CPPI originally requires no volatility estimation for its implementation, choosing the risk multiplier and the floor according to market conditions is necessary and the performance is mostly affected by market trends. An investor should therefore adjust the risk multiplier in CPPI to adapt to market conditions.

As a result, this paper proposes a new portfolio insurance strategy, named dynamic proportion portfolio insurance (DPPI), based on the risk variables which are used to build the equation tree for the risk multiplier by genetic programming. The performance of DPPI is compared to the traditional CPPI. In addition, principal component analysis is performed to determine the most important risk variables for the risk multiplier.

The rest of this paper is organized as follows. Section 2 reviews the research on constant proportion portfolio insurance (CPPI), genetic programming and principal component analysis. Section 3 describes our proposed dynamic proportion portfolio insurance (DPPI). The experimental process and experimental results are presented in Section 4. The concluding remarks and future directions are stated in Section 5.

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2. Background

2.1. Constant proportion portfolio insurance

Black and Jones (1987) introduced the constant proportion portfolio insurance (CPPI) strategy. The portfolio contains an active asset and a reserved risk-free asset. The active asset has a higher expected return than the reserved asset. For example, the active asset can be a stock while the reserved asset might be a T-bill. The CPPI strategy can be expressed by the following formula:

$$E = M \times (A - F),$$

where exposure E is the dollar amount invested in the active asset, asset A is the value of the portfolio, and floor F is the lowest acceptable value of the portfolio. The difference between asset and floor computes the cushion as the excess of the portfolio value over the floor. In addition, the risk multiplier M represents the ratio of the initial exposure to the initial cushion. When the active asset changes in value, the portfolio value will change accordingly and the cushion is also different from its initial value. The variable cushion is then multiplied by the constant risk multiplier, which is predetermined by the investor, to get the new exposure of the active asset.

This simple formula is easily understood by investors and helps them design their own strategies to meet their preferred level of risk. It is also straightforward to implement. However, determining the best risk multiplier to obtain a better investment result is a difficult task.

Genetic algorithms (GA) can be used to optimize the risk multiplier in CPPI in stead of using the investor's preference. However, the GA-optimized risk multiplier is still a constant which cannot adapt to the changing market conditions.

2.2. Genetic programming

Genetic Programming (Banzhaf, Nordin, Keller, & Francone, 1998; Koza, 1990, 1992, 1994a, 1994b) is a recent development which extends classical genetic algorithms (Bäck, Fogel, & Michalewicz, 2000a, 2000b; Goldberg, 1989; Holland, 1992; Mitchell, 1996) to process nonlinear structures. This optimization technique is based on the principles of natural evolution and is consisted of several genetic operators: selection, crossover, and mutation.

The major difference between genetic programming and genetic algorithms is the representation of the solution candidates. A hierarchical tree structure (as depicted in Fig. 1) represents a solution candidate in genetic programming while a string of characters with a fixed length represents a solution candidate in genetic algorithms.

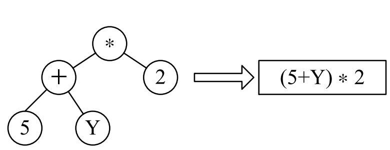

Fig. 1. A tree in genetic programming represents a formula.

The genetic programming framework consists of the flowing elements: node definition, initialization, fitness evaluation, selection, crossover, mutation, and termination condition.

Node definition: The nodes in the tree structure of genetic programming can be classified into two types. One of them is the terminal set which is consisted of constants or variables. The other one is the function set which is consisted of standard arithmetic operations, standard programming operations, standard mathematical functions, logical functions, or domain-specific functions. The elements of the terminal set and the function set are used to construct well-formed expression trees, which represent solutions to the problem, according to certain rules.

Initialization: Genetic programming starts with an initial population of expression trees which are randomly generated.

Fitness evaluation: Fitness evaluates how well a tree performs in the problem environment. Fitness values are used by the selection method to select trees for reproduction.

Selection: The selection method determines how trees are selected from the population to be parents for crossover. Better parents are usually selected with the hope that they have a better chance of producing better offsprings. The roulette wheel selection is a popular selection method.

Crossover: In GP, subtree crossover is the most common crossover operator. Subtree crossover works by replacing a subtree in one parent with a subtree from the other parent to produce the offspring. Fig. 2 presents an example of subtree crossover.

Mutation: In GP, mutation is usually achieved by replacing a subtree with a randomly generated subtree or by exchanging two randomly selected subtrees. An example of subtree replacement mutation is presented in Fig. 3.

Termination condition: Common termination conditions include fixed generations, fitness target, fitness convergence, and diversity convergence.

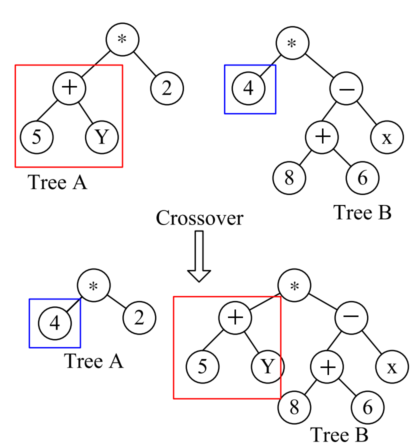

Fig. 2. An example of subtree crossover.

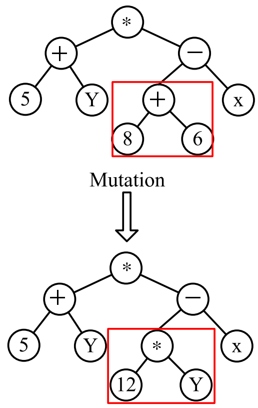

Fig. 3. An example of subtree replacement mutation.

2.3. Principal component analysis

Pearson (1901) developed the principal component analysis (PCA) technique. The purpose of principal component analysis is deducting the variables and extracting a few principal components to explain the relationships between variables.

PCA has been applied to many research domains which need to deal with a large amount of variables. For example, PCA has been used for feature selection (Guo, Wu, Massart, Boucon, & de Jong, 2002). PCA has also helped identify the meaning of the factor found in the portfolio analysis with CAPM (capital asset pricing model), and APT (arbitrage pricing theory) models (Merville, Hayes-Yelken, & Xu, 2001). It is a common statistical methodology for reducing variables and determining the their explanation powers.

3. Dynamic proportion portfolio insurance

Since the constant risk multiplier in CPPI cannot adapt to the changing market conditions, this study develops a new dynamical proportion portfolio insurance (DPPI) strategy whose risk multiplier is a function of risk variables generated by genetic programming.

The new DPPI strategy has a similar formula as in CPPI, but the risk multiplier in DPPI becomes a function of risk variables instead of a constant as in CPPI. The function is represented by an equation tree established by genetic programming. Genetic programming will pick up some risk variables to build the equation tree of the risk multiplier so that the resulting DPPI formula has the best performance in historical data. The formula represented by the equation tree will then calculate to different values of the risk multiplier at different future time points. The new DPPI formula becomes:

$$E = M(\mathscr{T}) \times (A - F),$$

where $\mathscr{T}$ is the set of constants and risk variables.

3.1. The terminal set

The terminal set of our genetic programming consists of constants and risk variables which include market volatility factors and technical indicators.

Market volatility factors: The book-to-market ratio has a strong positive relationship with the average stock return (Fama & French, 1992). When the beta of a portfolio is greater than one, there exists a positive correlation between market volatility and the portfolio (Schwert & Seguin, 1990). Since the beta is usually taken to be stable for some period of time, this study uses the risk-free rate and the market index instead of the beta in the capital asset pricing model (CAPM). Lastly, trading volume turnover rate of the whole market can explain the short-term volatility (Schwert, 1990), and the trading volume turnover rate of the individual stock is also considered in this study.

Technical indicators: In technical analysis, technical indicators have been used to predict the stock price variability (Levy, 1966). The stock price variability partly represents the market volatility. Thus, the technical indicators are also used in this study to construct the risk multiplier. The technical indicators used in several academic papers (Aby, Simpson, & Simpson, 1998; Brock, Lakonishok, & LeBaron, 1992; Kim & Han, 2000; Levy, 1967; Thawornwong, Enke, & Dagli, 2003; Tsaih, Hsu, & Lai, 1998) are listed in Table 1. The five technical indicators in Table 1 are considered in this study. However, since the moving average (MA) varies a lot from stock to stock, the disparity index (Nison, 1994), which represents the divergence of a price from its moving average, is substituted for the moving average. To summarize, the disparity index (DI), money flow index (MFI), relative strength index (RSI), stochastic oscillator (STO), and Williams %R (%R) are used in this study.

The risk variables used for the terminal set of our genetic programming are book-to-market ratio, risk-free rate, market index, exchange rate, market turnover rate, stock turnover rate, disparity index, relative strength index, stochastic oscillator, Williams %R, and money flow index. These risk variables and their abbreviations are listed in Table 2.

Since these risk variables have quite different value ranges, they are normalized to the same range before used in the equation trees.

Table 1
Technical indicators used in academic papers

Source	Technical indicators
Aby et al. (1998)	MFI, RSI
Brock et al. (1992)	MA
Kim and Han (2000)	RSI, STO, %R
Levy (1967)	RSI
Thawornwong et al. (2003)	MA, MFI, RSI, STO, %R
Tsaih et al. (1998)	RSI, STO

Table 2
The variables of the terminal set and their abbreviations

Variable	Abbreviation
Book-to-market ratio	B2M
Risk-free rate	T-bill (30-day)
Market index	NYSE (NYSE Composite)
Exchange rate	US2Euro (US dollar to Euro)
Market turnover rate	MT
Stock turnover rate	ST
Disparity index	DI
Money flow index	MFI
Relative strength index	RSI
Stochastic oscillator	STO-K & STO-D
Williams %R	%R

3.2. The function set

The function set of our genetic programming contains the operators for calculating the value of risk multiplier. The arithmetic operators commonly used in academic papers (Fyfe, Marney, & Tarbert, 1999; Kaboudan, 2000; Lensberg, 1999; Neely, Weller, & Dittmar, 1997; Potvin, Soriano, & Vallee, 2004) are listed in Table 3. This study adopts the four commonly used operators in Table 3: $+$, $-$, $\times$, $\div$.

3.3. Fitness

The fitness measure of our GP is

$$f = \frac{r+1}{\sigma+1},$$

where r rate of return after the transaction cost and tax are deducted and σ is the standard deviation. The addition of 1 to the rate of return is to avoid negative fitness and the addition of 1 to the standard deviation is to avoid zero denominator. To avoid unnecessary transaction costs due to frequent minor adjustments, the exposure is readjusted when the difference between the current exposure and the exposure calculated by the DPPI formula is greater than 5%.

To determine the contributions of the risk variables to the risk multiplier, principal component analysis is used to analyze their frequencies in the equation trees generated by genetic programming.

Table 3
Arithmetic operators used in genetic programming articles

Source	Operators
Fyfe et al. (1999)	$+, -, \times, \div$
Kaboudan (2000)	$+, -, \times, \div$, %, exp, ln, sqrt
Lensberg (1999)	$+, -, \times, \div$
Neely et al. (1997)	$+, -, \times, \div$
Potvin et al. (2004)	$+, -, \times, \div$

4. Experimental results

To show the effectiveness of our DPPI strategy, we compare it with the CPPI strategy whose constant risk multiplier is optimized by genetic algorithms. Their performances are then evaluated by paired t-test. This section presents the experimental process, the experimental results and the principal component analysis of the risk variables for the risk multiplier.

4.1. Data

In this study, five stocks from the New York Stock Exchange (NYSE) are selected for experimentation. The five stocks are IBM, LEE, MOT, UPS, and XOM which are described in Table 4.

The DPPI strategy produced by genetic programming based on the training period is evaluated on the following testing period. The six training and testing periods used for our experiments are listed in Table 5.

4.2. Parameters

The parameters of our genetic programming for DPPI are described in Table 6. The parameters of the genetic algorithm for CPPI are set to the same value whenever possible.

4.3. Results and analysis

Table 7 reports the fitness values of CPPI and DPPI on the six testing periods. The fitness value of DPPI is better than that of CPPI for the same company and testing period. The number of adjustment of DPPI is generally less than that of CPPI for the same company and testing period.

Table 4
The five companies used in this study

Symbol	Company	Industry
IBM	Int. Business machines	Computer hardware
LEE	Lee enterprises	Printing and publishing
MOT	Motorola	Comm. equipment
UPS	United parcel service	Trucking
XOM	Exxon mobil	Oil and Gas

Table 5
Training and testing periods of the experiments

Training period	Testing period
2001.07–12	2002.01–06
2002.01–06	2002.07–12
2002.07–12	2003.01–06
2003.01–06	2003.07–12
2003.07–12	2004.01–06
2004.01–06	2004.06–12

Table 6
Parameters of our genetic programming runs

Parameter	Value
Population size	500
Termination condition	200 generations
Maximum tree depth	8
Selection method	Roulette wheel
Crossover method	Subtree crossover
Crossover rate	100%↘15%
Mutation method	Subtree replacement
Mutation rate	1%
Elitism	Top two of a generation

Table 7
Performances of CPPI and DPPI on testing periods

Period	Stock	CPPI		DPPI	
		Fitness	Adj.	Fitness	Adj.
2002.01–06	IBM	0.8608	111	0.9257	33
	LEE	0.8011	29	0.9470	35
	MOT	0.8647	88	1.0005	1
	UPS	0.8474	87	0.9057	12
	XOM	0.7926	36	1.0183	2
2002.07–12	IBM	0.8511	93	0.9452	48
	LEE	0.8419	79	0.9471	92
	MOT	0.8410	98	0.9348	97
	UPS	0.5995	1	0.9930	7
	XOM	0.8363	113	0.9069	11
2003.01–06	IBM	0.6293	3	0.9119	16
	LEE	0.8223	57	1.0113	3
	MOT	0.8636	84	0.9405	46
	UPS	0.8434	61	0.9443	61
	XOM	0.7554	21	0.9465	18
2003.07–12	IBM	0.5997	1	1.0003	1
	LEE	0.7470	28	0.9982	5
	MOT	0.6602	5	0.9821	8
	UPS	0.8189	54	0.9994	13
	XOM	0.7825	31	0.9945	31
2004.01–06	IBM	0.7531	20	0.9276	12
	LEE	0.6003	1	0.9726	7
	MOT	0.6072	1	0.9908	6
	UPS	0.8087	36	0.9872	25
	XOM	0.5997	1	0.9628	18
2004.07–12	IBM	0.8139	36	1.0075	3
	LEE	0.7464	16	0.9524	11
	MOT	0.8426	94	0.998	3
	UPS	0.8201	55	0.9935	55
	XOM	0.5996	1	1.0085	4
Average		0.7622	45.28	0.9685	22.2
Std. dev.		0.012		0.0015	

The result of paired t-test on fitness values of CPPI and DPPI is given in Table 8. According to the paired t-test result, DPPI significantly outperforms GA-optimzed CPPI. The result shows that dynamic risk multiplier of the risk variables really works better than the constant risk multiplier determined by genetic algorithms or by the investor's preference.

Table 9 shows the principal component analysis results of the leading risk variables for the risk multiplier. The result reveals that the risk-free rate (T-bill), influences the risk multiplier the most, followed in turn by stochastic D, the market index (NYSE), and the exchange rate (US2Euro). Among the top four risk variables, risk-free rate, market index, and exchange rate belong to the market volatility factors, while stochastic D belongs to the individual stock's technical indicators. Furthermore, one market volatility factor (book-to-market ratio) and two of the technical indicators (disparity index and relative strength index) have little influence on the risk multiplier.

The top six risk variables of yearly principal component analysis are given in Table 10 where three risk variables, stochastic D, exchange rate (T-bill), and market index (NYSE), repeatedly appear in each year with different orders.

Table 8
The result of paired t test on fitness values of CPPI and DPPI

H_0:	$\mu_D - \mu_C \leqslant 0$
H_1:	$\mu_D - \mu_C > 0$
D:	DPPI
C:	CPPI
Decision: Reject H_0 if $t > t_{(0.95,58)} = 1.6716$	
Result: $t = 10.8692 > 1.6716$, H_0 is rejected	

Table 9
PCA results of the leading risk variables for the risk multiplier

Variable	PCA value
T-bill	0.817
STO-D	0.709
NYSE	0.686
US2Euro	0.649
ST	0.544
%R	0.507
MT	0.446
STO-K	0.437
MFI	0.359

Table 10
PCA results by year

	2002	2003	2004
1	STO-D	T-bill	NYSE
2	T-bill	US2Euro	T-bill
3	US2Euro	MT	STO-D
4	ST	STO-D	%R
5	NYSE	MFI	STO-K
6	%R	NYSE	ST

Table 11
PCA results by company

	IBM	LEE	MOT	UPS	XOM
1	T-bill	T-bill	STO-D	ST	STO-D
2	US2Euro	STO-D	NYSE	MT	ST
3	NYSE	%R	US2Euro	T-bill	%R
4	MFI	ST	T-bill	%R	US2Euro
5	%R	NYSE	MT	NYSE	MFI
6	MT	US2Euro	STO-K	US2Euro	T-bill

Table 11 lists the PCA results for each company. Three risk variables, T-bill, exchange rate, and NYSE, repeatedly appear in each company with different order.

5. Conclusion

This paper proposes a dynamic proportion portfolio insurance (DPPI) strategy which generalizes the well-known CPPI strategy by making the constant risk multiplier in CPPI a function of risk variables. It allocates capital in the risky asset by dynamically adjusting the risk multiplier according to those risk variables. The risk multiplier function, represented as an equation tree, is generated using genetic programming. The experimental results reveal that our DPPI significantly outperforms the traditional CPPI.

In addition, principal component analysis is used to determine the leading risk variables in the equation trees generated by genetic programming. The principal component analysis results show that some of the risk variables used in our experiments do appear more often than others.

Future works include finding more risk variables and using automatically defined functions in genetic programming to simplify the equation trees. Another interesting direction is extending our DPPI idea to time invariant portfolio protection (TIPP) (Estep & Kritzman, 1988) which can be expressed by the following formulas:

$$E_{t+1} = M \times (A_t - F_t),$$
$$F_t = \lambda \times \max_{\tau \leqslant t} A_\tau,$$
$$\lambda = F_0 / A_0.$$

By making the risk multiplier in TIPP a function as in DPPI, a dynamic TIPP strategy with a variable risk multiplier can be constructed.

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